

DATA ANALYSIS PROJECT

Route & Cost Optimization Simulation for Rice Aid Distribution

Case Study: PT Pos Indonesia, Regional Office III Bandung —
West Java

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— PROBLEM CONTEXT

Scale of the Distribution Network

The Regional Office III Bandung rice aid program spans dozens of warehouses and thousands of distribution points across West Java — a route network complex enough to warrant optimization.

38

Bulog Warehouses (Origins)

3,327

Village-Level Distribution Points

22

Regencies/Cities in West Java

±36,454 tons

Total Rice Allocation

DATA ISSUE IDENTIFIED

Transport Tariff Column Was Empty

The original dataset had no transport cost or tariff data filled in. Rather than a dead end, this gap became the entry point for analysis: a documented cost simulation model built to practice data analysis skills.

Data gap → entry point for analysis, not a dead end.

— METHODOLOGY

Two-Step Simulation

01 Route Consolidation

Merged shipments sharing the same origin and destination, recalculating trip counts from total allocation — reducing trucks dispatched below capacity.

DATA-DRIVEN

02 Additional Route Efficiency Simulation

Multi-drop routing between nearby points and reallocation to the nearest warehouse, using documented assumption parameters.

DOCUMENTED ASSUMPTION

— BEFORE VS AFTER

Total Midlemile Cost

Before

IDR 6.67B



After

IDR 4.47B

-33%

COST REDUCTION

SUPPORTING METRICS

Route & Trip Consolidation

Total Routes

Before

8,591

After

4,869



Total Trips

Before

11,355

After

9,710



— TOOLS & SKILLS

Applied on This Project

- **Python**

pandas, openpyxl for data cleaning and simulation

- **Excel**

Live formulas (not hardcoded), dashboard workbook

- **Data Cleaning**

Comma-decimal parsing, forward-filling missing values

- **Assumption Documentation**

Every simulation parameter logged and traceable

— KEY TAKEAWAY

A Data Gap Isn't the End of Analysis — It's the Start

What began as an empty tariff column grew into a route optimization simulation that cut midmile costs by 33%, fully documented and auditable.

Thank you for reading

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